

Question	Answer	Mark	Mark scheme	Additional guidance
16	15	P1	for process to find volume of cylinder or volume of cone, eg $\pi \times 8^2 \times \frac{h}{3}$ oe or $\frac{1}{3} \times \pi \times 8^2 \times h$ oe or $\pi \times 8^2 \times H$ oe or $\frac{1}{3} \times \pi \times 8^2 \times 3H$ oe or $\pi \times 8^2 \times \frac{x}{4}$ oe or $\frac{1}{3} \times \pi \times 8^2 \times \frac{3x}{4}$ oe	h = height of the cone H = height of the cylinder x = total height of the shape Allow any letter for h, H and x, does not have to be defined for the award of the marks The award of all marks requires the substitution of $r = 8$, allow this to be done at a later stage in the question
		P1	(dep P1) for setting up an equation in terms of one variable, eg "$\pi \times 8^2 \times \frac{h}{3}$" + "$\frac{1}{3} \times \pi \times 8^2 \times h$" = 640π oe or "$\pi \times 8^2 \times H$" + "$\frac{1}{3} \times \pi \times 8^2 \times 3H$" = 640π oe or "$\pi \times 8^2 \times \frac{x}{4}$" + "$\frac{1}{3} \times \pi \times 8^2 \times \frac{3x}{4}$" = 640π oe or "$\pi \times 8^2 \times \frac{h}{3}$" = 320π oe or "$\frac{1}{3} \times \pi \times 8^2 \times h$" = 320π oe or "$\pi \times 8^2 \times H$" = 320π oe or "$\frac{1}{3} \times \pi \times 8^2 \times 3H$" = 320π oe or "$\pi \times 8^2 \times \frac{x}{4}$" = 320π oe or "$\frac{1}{3} \times \pi \times 8^2 \times \frac{3x}{4}$" = 320π oe	A correct equation implies the first P1 Allow inconsistent use of π within their equation provided the correct volumes are seen previously

Question	Answer	Mark	Mark scheme	Additional guidance
		P1	(dep P2) for process to solve for h or H or x eg $(h =) \frac{640\pi}{\frac{1}{3} \times \pi \times 8^2 + \frac{1}{3} \times \pi \times 8^2}$ oe eg $(h =) \frac{3 \times 640\pi}{\text{"64}\pi" + \text{"64}\pi"}$ or $(H =) \frac{640\pi}{\pi \times 8^2 + \frac{1}{3} \times \pi \times 8^2 \times 3} (= 5)$ oe eg $(h =) \frac{640\pi}{\text{"64}\pi" + \text{"64}\pi"} (= 5)$ or $(x =) \frac{640\pi}{\pi \times 8^2 \times \frac{1}{4} + \frac{1}{3} \times \pi \times 8^2 \times \frac{3}{4}} (= 20)$ oe eg $(x =) \frac{640\pi}{\text{"32}\pi"} (= 20)$	Can be an equation in the form $ah = p$ or $bH = q$ or $cx = r$ where a or b or c is an integer
		A1	cao	Award 0 marks for a correct answer with no (or incorrect) supportive working